



XMBA 201A
Managerial Economics

Decision Trees with Probabilistic Events (with applications to information value and option value)

Lecture 2

Professor Steve Tadelis

1

Decision Trees: Recap

- Decision trees help **FRAME PROBLEMS**:
 - Boil things down **to core issues**
 - Force **critical thinking**
 - Recognize/debate **alternative proposals and implicit assumptions**
- \$64K question: values and probabilities?
 - Use **all relevant information** and **cost effective experimentation**
 - Must **hold people accountable** (incentives matter!!)
 - Consider **risk attitudes and unknown probabilities** (Numbers...)
- **Don't confuse good/bad outcomes with good/bad decisions!**

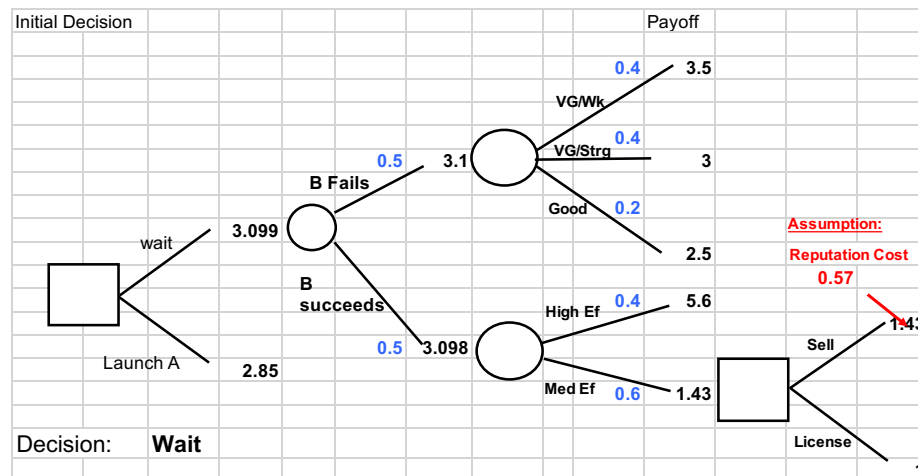
2

Merck Pharma and Covid Vaccines

1. What is the order of events and the decisions to be made? Draw the decision tree for the case.
2. What judgment calls do you need to make in constructing the decision tree? That is, what is left somewhat (or more than somewhat) ambiguous in the description?
3. How much would it be worth to know ahead of time if version B will succeed?
4. Would the decision maker's degree of risk aversion possibly change their choices?

3

The Decision Problem:



Value of knowing if B will succeed? 1 cent loss per vaccine saved, but only with probability 0.5 ! ($0.5 \times 0.01 \times Q$)

Risk aversion? Makes waiting less desirable because of the risk from low outcomes

4

Risk attitudes: definitions

- Consider a lottery L that pays \$0 or \$100 with equal probabilities
- A **risk-neutral** decision maker: lottery is as good as its expected value:
 - $Ev(L) = v(\$50)$ (e.g., $v(x) = x$ for any value x)
- A **risk-averse** decision maker: lottery is inferior to its expected value:
 - $Ev(L) < v(\$50)$ (e.g., $v(x) = \sqrt{x}$ for any value x)
- A **risk-loving** decision maker: lottery is superior to its expected value:
 - $Ev(L) > v(\$50)$ (e.g., $v(x) = x^2$ for any value x)

5

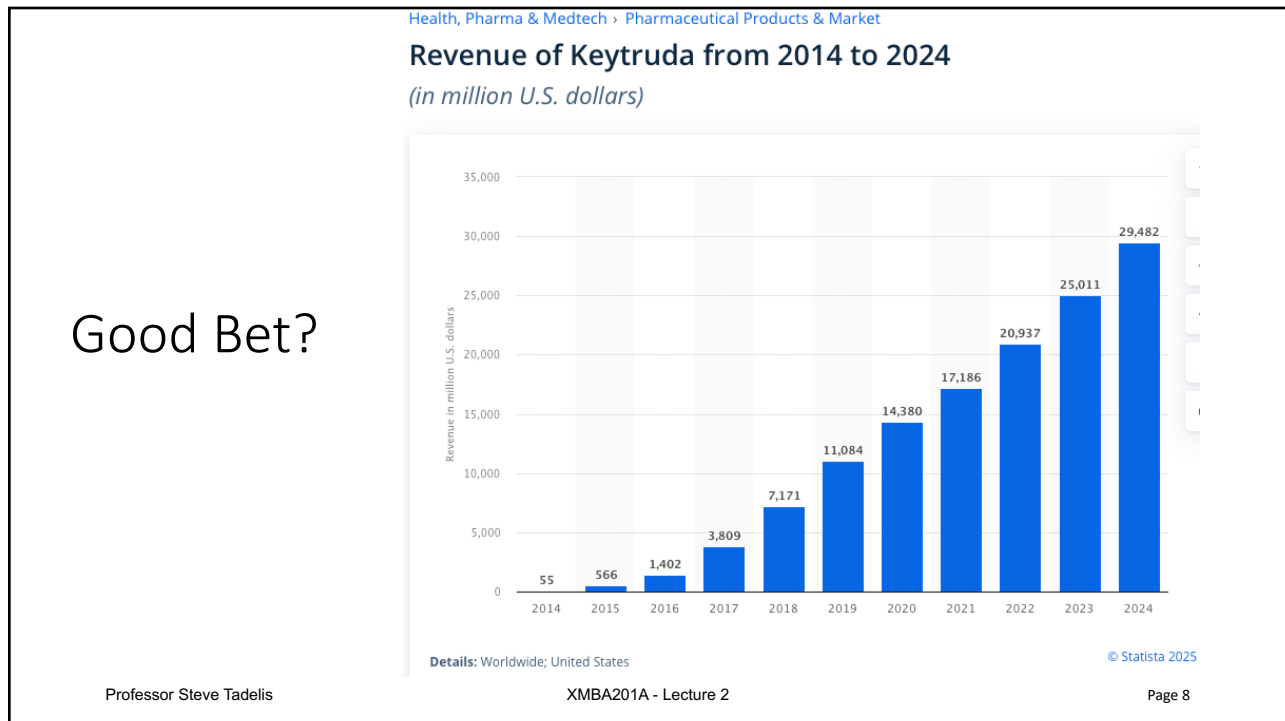
Why Merck Is Betting Big on One Cancer Drug

1. Why is Merck shifting R&D resources to its Keytruda drug?
2. Suppose a risk-neutral monopoly pharmaceutical company has two drugs on which it can perform R&D. The total R&D time that can be devoted to the drugs is normalized to 1, with r_i denoting the amount of time devoted to drug i so that $r_1 + r_2 = 1$. Each drug is either a success, which has payoff x_i for drug i . The probabilities of success, $p_i(r_i)$, is a function of the amount of time r_i devoted to drug i . How do you think that the optimal R&D time devoted to the two drugs depends on the probability of success functions $p_1(r_1)$ and $p_2(r_2)$? (Think especially about how their shape affects the results.)
3. How would a risk-averse drug company allocate R&D resources between the two drugs compared to a risk-neutral one?

6



7



8

Real Options and Option Value

Professor Steve Tadelis

XMBA201A - Lecture 2

Page 9

9

Real Options and Reversibility

- Imagine that you have a proprietary family lemonade recipe that guaranties you \$100 of profit per day, and that you can operate it today and tomorrow, after which you dedicate yourself to studies!
- Your total profit will therefore be $\$100 + \$100 = \$200$.
- Now imagine that you can try an experimental recipe that is expected to generate a total daily profit of **either** $\$(x + 50)$ **or** $\$(x - 50)$, with equal probability, so the expected daily profit is x .
- ***What is the lowest value of x that makes the experiment worthwhile?***
- Intuitively, you'd think it's $x = \$100$ because then on expectations you'll get \$100 a day: $(\frac{1}{2} \cdot 150 + \frac{1}{2} \cdot 50 = 100)$. ***But can it be less?***

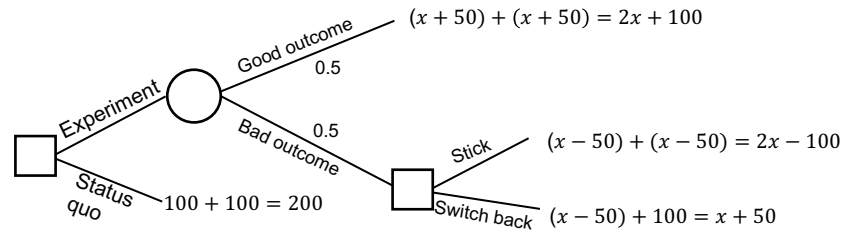
Professor Steve Tadelis

XMBA201A - Lecture 2

Page 10

10

Real Options and Reversibility: $50 < x < 100$



- Switching is valuable if and only if:

$$\frac{1}{2} \cdot (2x + 100) + \frac{1}{2} \cdot (x + 50) > 200, \quad \text{or } x > 83\frac{1}{3}$$

- Switching is **reversible**: downside is bounded so upside can be smaller than what you think you need!

11

Pricing the Option of Reversibility

- Imagine that $x = \$90$ so the good outcome yields \$140 while the bad yields \$40. With the option, the expected value of experimenting is:

$$E[\text{profit}] = \frac{1}{2} \cdot (2x + 100) + \frac{1}{2} \cdot (x + 50) = \frac{1}{2} \cdot 280 + \frac{1}{2} \cdot 140 = \$210$$

- Now imagine that your classmate wants to buy your proprietary recipe before you try the experimental one, after which you will not be able to switch back. **For how much should you sell it?**
- Without the option to switch back, expected value of experimenting is:

$$E[\text{Profit}] = \frac{1}{2} \cdot (2x + 100) + \frac{1}{2} \cdot (2x - 100) = \frac{1}{2} \cdot 280 + \frac{1}{2} \cdot 80 = \$180$$

- How much is the option to switch back worth?

$$\text{Option Value} = 210 - 180 = \$30$$

12

A Primer on Discounted Cash Flows

- Imagine that you are promised \$100 in exactly 1 year. How much is that worth for you today? (What would you trade that promise for today?)
- ***How much can you borrow today from a bank if you give them that promise note?***

13

WELLS FARGO

[Sign On](#)
[Customer Service](#)
[ATMs/Locations](#)
[Español](#)

[Personal](#)
[Investing & Wealth Management](#)
[Small Business](#)
[Commercial Banking](#)
[Corporate & Investment Banking](#)

[Personal](#) > [Personal Loans](#) > [Personal Loan Rates](#)

Personal Loan Rates

[Print](#)
[Share](#)



Personal loan interest rates as low as 7.49% APR^{1,2}

For a \$10,000 Wells Fargo Personal Loan with a 3-year term.²

Get started

The Annual Percentage Rate (APR) shown is for a personal loan of at least \$10,000, with a 3-year term and includes a relationship discount of 0.25%.^{1,3} Your actual APR may be higher than the rate shown.

To be eligible for a personal loan, you are required to have an open Wells Fargo account for at least 12 months.

Personal Loan Loan amounts from \$3,000 to \$100,000 with no origination fee or prepayment penalty. See my loan options >	Fixed rates Fixed rates range from 7.49% to 24.99% APR^{2,5} APR includes a 0.25% relationship discount.	Quick credit decision Same-day credit decision for most customers.
--	--	--

Helpful Links

- [Learn the benefits of loans](#)
- [What information do I need to apply?](#)
- [Find answers to common questions](#)

Still have questions?

14

A Primer on Discounted Cash Flows

- Imagine that you are promised \$1,000 in exactly 1 year. How much is that worth for you today? (What would you trade that promise for today?)
- **How much can you borrow today from a bank if you give them that promise note?**
Imagine they charge you an interest rate of 12%.
 - ⇒ For every \$1 you borrow today, you must return \$1.12 in a year
 - ⇒ If you borrow \$x today, you must return \$y = \$x × 1.12 in a year
 - ⇒ If you can pay back \$y in a year, the bank will lend you $\$x = \frac{y}{1.12}$ today!
- **Conclusion:** \$1,000 in one year is worth $\frac{1,000}{1.12} = 892.86$ today!
- What if you'll get the \$1,000 in two years? You multiply twice!
 - ⇒ If you borrow \$x today, you must return \$y = \$x × 1.12 × 1.12 in 2 years
 - ⇒ If you pay back \$y in 2 years, the bank will lend you $\$x = \frac{y}{(1.12)^2}$ today!

Professor Steve Tadelis

XMBA201A - Lecture 2

Page 15

15

Real Options: Harder example (self study)

- Imagine that you have a proprietary production procedure that you use for some process.
- The profits from are \$2M/yr for the next 5 years, and and Internal Rate of Return (IRR) = 10% (This is the company's "value of future money")
 - $NPV = 2 + \frac{2}{1+r} + \frac{2}{(1+r)^2} + \dots + \frac{2}{(1+r)^4} = \$8.34M$
- You can replace your procedure with an experimental one that is expected to generate a total yearly profit of $\$(x + 1)$ or $\$(x - 1)$, each with equal probability, so the expected yearly profit is x.
- What is the lowest value of x that would make the switch worthwhile?

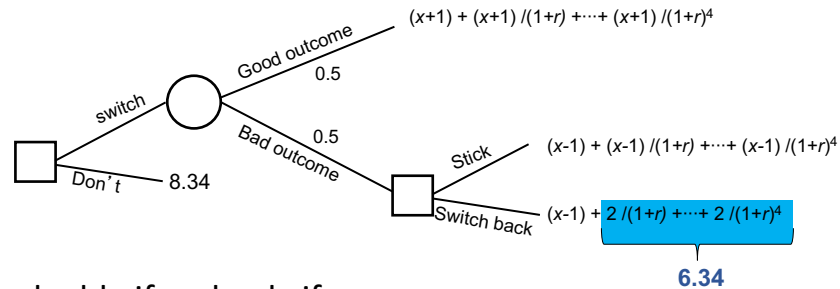
Professor Steve Tadelis

XMBA201A - Lecture 2

Page 16

16

Real Options and Reversibility



- Switching is valuable if and only if
- $\frac{1}{2}(x - 1 + 6.34) + \frac{1}{2}(x + 1)[1 + 1/(1 + r) + \dots + 1/(1+r)^4] > 8.34$
or, $x > 1.39M$!
- Switching is **reversible**: downside is bounded so upside can be smaller than a first glance suggests!

Professor Steve Tadelis

XMBA201A - Lecture 2

Page 17

17

Pricing the Option of Reversibility

- Imagine that $x = \$1.5M$ so the good outcome yields \$2.5M while the bad yields \$0.5M. With the option, the expected value of switching is:

$$NPV = \frac{1}{2} \times 2.5 \times \left(1 + \frac{1}{1+r} + \dots + \frac{1}{(1+r)^4}\right) + \frac{1}{2} \times (0.5 + 6.34) = \$8.63M$$

- Now imagine another firm wants to buy your proprietary process before you try the experimental one, after which you will not be able to switch back. Should you sell it?

- Without the option to switch back, expected value of switching is:

$$NPV = \frac{1}{2} \times 2.5 \times \left(1 + \frac{1}{1+r} + \dots + \frac{1}{(1+r)^4}\right) + \frac{1}{2} \times 0.5 \times \left(1 + \frac{1}{1+r} + \dots + \frac{1}{(1+r)^4}\right) = \$6.25M$$

- How much is the option to switch worth?

$$\text{Option Value} = 8.63 - 6.25 = \$2.38M$$

Professor Steve Tadelis

XMBA201A - Lecture 2

Page 18

18

Information & Real Options: Summary

- Acquiring information adds ***contingent decisions*** to the tree
 - Each tree results in a ***best expected value***
 - The difference in expected values across the trees is the ***value of information***
- Having an option adds ***contingent decisions*** to the tree
 - Each tree results in a ***best expected value***
 - The difference in expected values across the trees is the ***value of the option***
- Decision Tree analysis is a very versatile tool!

19

Introduction to Economic Costs

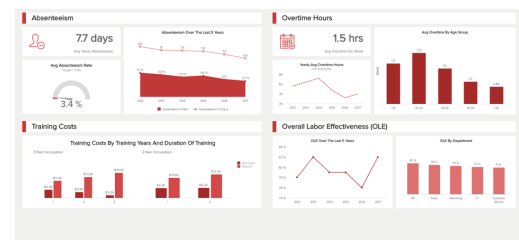
20

Make or Buy?

- Developing a new HR dashboard is valued at \$1M (reduces churn).
- An external developer bid comes in at \$250K.
- You can develop it internally: time and cloud services
 - The cloud services cost \$75K on the market
 - Need 1,800 engineering hours to build it.
- Your hourly engineering wage is \$125 all inclusive.
- What should you do?
- Doing it yourself:

$$\$75K + \$125 \times 1800 = \$300K$$

→ **Buying from developer is better!**



21

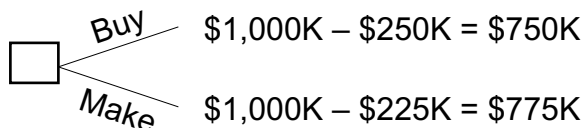
The Friendly Cloud Provider

- You have not yet hired the external developer
- An old friend runs a cloud computing company and offers you the cloud services you need for this project **for free!**
- Realizing that you just got the cloud services for free, **what should you do?**

22

You're an accountant:

- Hiring the outside developer costs \$250K.
- Doing it yourself costs only your labor (the cloud is free!)
 - Cost = $125 \times 1800 = \$225K$
- The accountant's decision tree:

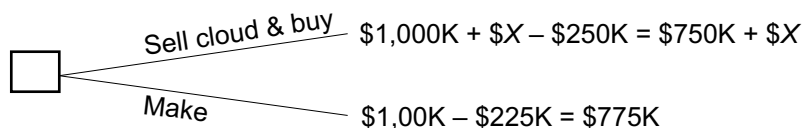


- **Conclusion:** do it yourself!

23

You're an Economist:

- Hiring the outside developer costs \$250K.
- Doing it yourself may seem to take only your time
- You can "sell" the cloud for \$X! The **opportunity cost** to use it is \$X!
 - Or use the free cloud for another activity that you're currently paying for that's worth \$X...
- The Economist's decision tree:

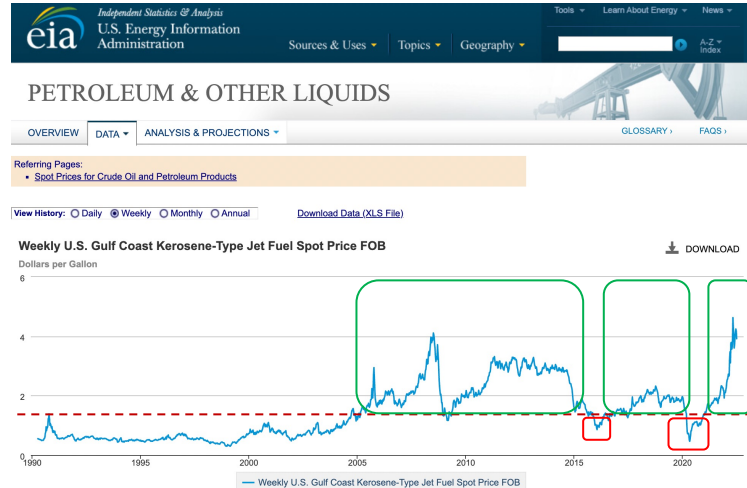


- **Conclusion:** If $X > \$25K$ then sell the cloud capacity and hire developer!
 - Or use the cloud access to bargain down the developer by at least \$25K.

24

Airline Fuel Hedging

- If jet fuel is trading at \$1.5/gallon, you can hedge possible price increases by purchasing a financial option that allows you to buy X gallons at \$1.5/gl in the future.
 - If price falls *below* \$1.5/gln, option is worthless. They buy fuel at the lower spot price.
 - If price goes *above* \$1.5/gln, they can purchase up to X gallons at \$1.5/gln.



Professor Steve Tadelis

XMBA201A - Lecture 2

Page 25

25

Jet Fuel Options: Some old headlines

- **Goldman Upgrades Southwest Air On Fuel-Hedging Strength (DJ 9/2/05)**
- **Southwest Flies Above Industry Challenges With Focus on Efficiency (Yahoo Finance, 8/23/06)**
- **Southwest Airlines (LUV) Thursday reports what almost certainly will be its 69th-consecutive quarterly profit, a string dating back to 1991. It won't be the first time that the USA's leading discount carrier has earned money while a financial crisis threatened other carriers' survival. What is its secret? There are lots of factors, but easily the most important is that Southwest is the champion oil price hedger among airlines worldwide. (USA Today, 7/24/08)**



Professor Steve Tadelis

XMBA201A - Lecture 2

Page 26

26

Airline jet fuel hedge: a toy example

- Imagine that Southwest has a route for which it breaks even at \$1.51/gallon (ignore reputation, etc.). In other words, the route is only profitable if the jet fuel price is below \$1.51/gln.
- Should Southwest's decision to fly that route depend on whether or not it has hedged its fuel costs?
- Data:
 - Current price is \$1.5/gln
 - Prices may either increase to \$2/gln or drop to \$1/gln
 - Each event is equally likely
 - How does holding hedges affect decisions?

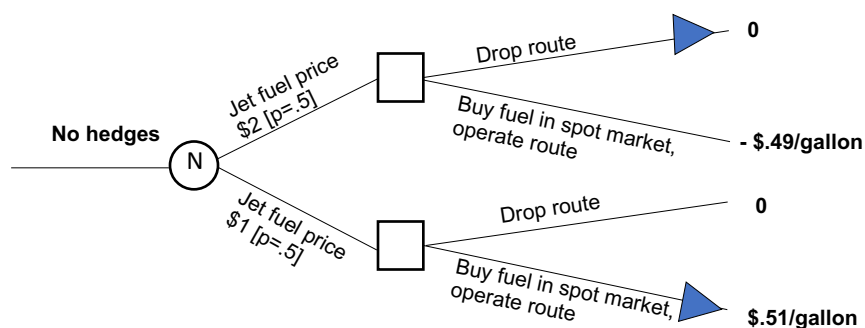
Professor Steve Tadelis

XMBA201A - Lecture 2

Page 27

27

Southwest's decision if it's not-hedged



What changes when they **start with options to buy fuel at \$1.50?**

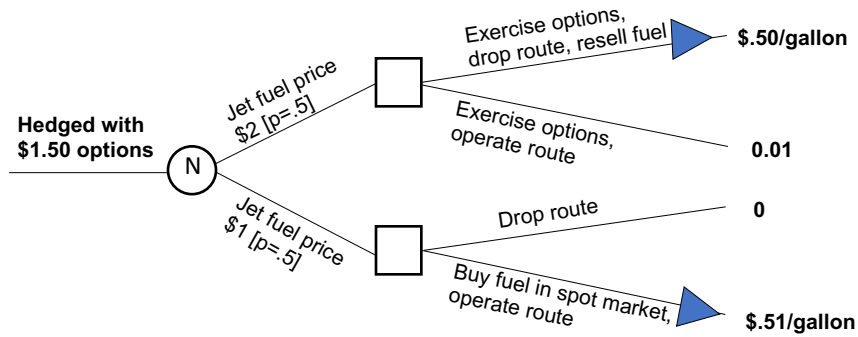
Professor Steve Tadelis

XMBA201A - Lecture 2

Page 28

28

Southwest's decision if it's hedged



Result: Although SWA's **acquisition cost** depends on whether it has hedges, its **opportunity cost** reflects the **spot price** in either case.

29

Farms in Downtown Tokyo



30

Opportunity cost vs. Expenses

- What are The costs to get a FULL TIME MBA?

	Cost	Expense	
	Tuition	Tuition	
	Books, etc.	Books, etc.	
Opportunity Cost, not Expense!	Salary	—	
	—	Room & Board	Expense, Not Opportunity Cost!

- Some costs are not expenses (salary loss) and vice versa (R&B)
- Is it the same for EMBA students?

31

Back to HR Dashboard

- Recall:
 - An external developer bid would be \$250K.
 - You can develop it internally: time and cloud services
 - The cloud services cost \$75K on the market
 - Need 1,800 engineering hours to build it.
 - Hourly engineering wage is \$125 all inclusive
- $\Rightarrow costs = 75,000 + 1800 \times 125 = \$300,000$
- $\Rightarrow$ You should hire the developer!
- Imagine you chose to build yourself not aware of a qualified outside developer.
- After purchasing cloud for \$75K **but not using it** and putting in **200 hours** of R&D engineering (at \$125/hr = \$25K), you learn about the option of hiring a developer company at \$250K.
- You only need 1,600 hours of engineering left (\$200K). **What should you do?**

32

Back to HR Dashboard

- If you can get at least \$50K for the cloud capacity remaining, then you should hire the contractor!
- The \$25K of labor are a **sunk cost!** They have been “burned”, and do not affect any of the future alternatives! Therefore, they should be ignored!

33

Sunk Cost fallacy: Examples

- The non-recoverable portion of irreversible expenditures made in the past:
 - ***What does this imply about project champions?***
(don't confuse good/bad outcomes with good/bad decisions...)
- The price of the movie ticket for a bad movie.
- The cost paid for a vacation/hobby/training in advance
- **The sunk-cost fallacy has a powerful pull, and it takes lots of discipline to resist it!**

34

Economic Costs vs. Accounting Costs: Summary

- There are two things to do with expenses:
 - Keep track of them (EXPENSES: RECORD KEEPING).
 - Use them for decisions (COSTS: DECISION MAKING).
- Accounting rules are designed to help firms & investors keep track of **expenses**: where did the money go? Where did it come from?
- A **first key difference** between economists and accountants is that accountants are less able to recognize opportunity costs.
- A **second key difference** between economists and accountants is that accountants are not allowed to ignore sunk costs.
- Separating accounting objectives from decision objectives is critical!!